Chapter 19 : Chemical Coordination and Integration

1. Match List-I with List-II. **[NEET-2025]**

List-I	List-II
A. Progesterone	I. Pars intermedia
B. Relaxin	II. Ovary
C. Melanocyte stimulating hormone	III. Adrenal Medulla
D. Catecholamines	IV. Corpus luteum

Choose the correct answer from the options given below :

(1) A-IV, B-II, C-I, D-III

(2) A-IV, B-II, C-III, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-III, B-II, C-IV, D-I

2. Consider the following statements regarding function of adrenal medullary hormones : **[NEET-2025]**

(A) It causes pupilary constriction.

(B) It is a hyperglycemic hormone.

(C) It causes piloerection.

(D) It increases strength of heart contraction.

Choose the correct answer from the options given below :

(1) C and D only

(2) B, C and D only

(3) A, C and D only

(4) D only

3. Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus?

(1) Luteinizing hormone (LH) **[NEET-2025]**

(2) Anti-diuretic hormone (ADH)

(3) Follicle-stimulating hormone (FSH)

(4) Adrenocorticotropic hormone (ACTH)

4. Why can't insulin be given orally to diabetic patients? **[NEET-2025]**

(1) Human body will elicit strong immune response

(2) It will be digested in Gastro-Intestinal (GI) tract

(3) Because of structural variation

(4) Its bioavailability will be increased

5. Match List-I with List-II. **[NEET-2025]**

List-I	List-II
A. Heart	I. Erythropoietin
B. Kidney	II. Aldosterone
C. Gastro-intestinal tract	III. Atrial natriuretic factor
D. Adrenal Cortex	IV. Secretin

Choose the correct answer from the options given below :

(1) A-II, B-I, C-III, D-IV (2) A-IV, B-III, C-II, D-I

(3) A-I, B-III, C-IV, D-II (4) A-III, B-I, C-IV, D-II

6. Match List-I with List-II. **[Re-NEET-2024]**

List-I	List-II
A. Epinephrine	I. Hyperglycemia
B. Thyroxine	II. Smooth muscle contraction
C. Oxytocin	III. Basal metabolic rate
D. Glucagon	IV. Emergency hormone

Choose the correct answer from the options given below :

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-I, B-IV, C-III, D-II

7. Identify the wrong statements : **[Re-NEET-2024]**

A. Erythropoietin is produced by juxtaglomerular cells of the kidney

B. Leydig cells produce Androgens

C. Atrial Natriuretic factor, a peptide hormone is secreted by the seminiferous tubules of the testes

D. Cholecystokinin is produced by gastrointestinal tract

E. Gastrin acts on intestinal wall and helps in the production of pepsinogen

Choose the most appropriate answer from the options given below :

(1) D and E only (2) A and B only

(3) C and E only (4) A and C only

8. Which of the following is not a steroid hormone? **[NEET-2024]**

(1) Cortisol

(2) Testosterone

(3) Progesterone

(4) Glucagon

9. Match List I with List II : **[NEET-2024]**

List I	List II
A. Exophthalmic goiter	I. Excess secretion of cortisol, moon face & hypergylcemia.
B. Acromegaly	II. Hypo-secretion of thyroid hormone and stunted growth.
C. Cushing's syndrome	III. Hyper secretion of thyroid hormone & protruding eye balls.
D. Cretinism	IV. Excessive secretion of growth hormone.

Choose the correct answer from the options given below :

(1) A-I, B-III, C-II, D-IV

(2) A-IV, B-II, C-I, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-IV, C-I, D-II

10. Which of the following statements are correct with respect to the hormone and its function?

(A) Thyrocalcitonin (TCT) regulates the blood calcium level. **[NEET-2023-MANIPUR]**

(B) In males, FSH and androgens regulate spermatogenesis.

(C) Hyperthyroidism can lead to goitre.

(D) Glucocorticoids are secreted in Adrenal Medulla.

(E) Parathyroid hormone is regulated by circulating levels of sodium ions.

Choose the most appropriate answer from the options given below :

(1) (B) and (C) only (2) (A) and (D) only

(3) (C) and (E) only (4) (A) and (B) only

11. Given below are two statements: **[NEET-2023-MANIPUR]**

Statement I : Parathyroid hormone acts on bones and stimulates the process of bone resorption.

Statements II : Parathyroid hormone along with Thyrocalcitonin plays a significant role in carbohydrate metabolism.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is correct but Statement II is false

(4) Statement I is incorrect but Statement II is true

12. Match List I with List II. **[NEET-2023]**

List I	List II
A. CCK	I. Kidney
B. GIP	II. Heart
C. ANF	III. Gastric gland
D. ADH	IV. Pancreas

Choose the correct answer from the options given below :

(1)A-IV, B-III, C-II, D-I

(2) A-III, B-II, C-IV, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-II, C-III, D-I

13. Which of the following statements are correct? **[NEET-2023]**

A. An excessive loss of body fluid from the body switches off osmoreceptors.

B. ADH facilitates water reabsorption to prevent diuresis.

C. ANF causes vasodilation.

D. ADH causes increase in blood pressure.

E. ADH is responsible for decrease in GFR.

Choose the correct answer from the options given below:

(1) A and B only

(2) B, C and D only

(3) A, B and E only

(4) C, D and E only

14. Which of the following are NOT under the control of thyroid hormone? **[NEET-2023]**

A. Maintenance of water and electrolyte balance

B. Regulation of basal metabolic rate

C. Normal rhythm of sleep-wake cycle

D. Development of immune system

E. Support the process of RBCs formation

Choose the correct answer from the options given below:

(1) A and D only

(2) B and C only

(3) C and D only

(4) D and E only

15. Match List -I with List -II. **[NEET-2022]**

List-I (Biological Molecules)	List-II (Biological functions)
(a) Glycogen	(i) Hormone
(b) Globulin	(ii) Biocatalyst
(c) Steroids	(iii) Antibody
(d) Thrombin	(iv) Storage product

Choose the correct answer from the options given below:

(1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

16. Which of the following are not the effects of Parathyroid hormone ? **[NEET-2022]**

(a) Stimulates the process of bone resorption

(b) Decreases Ca2+ level in blood

(c) Reabsorption of Ca2+ by renal tubules

(d) Decreases the absorption of Ca2+ from digested food

(e) Increases metabolism of carbohydrates

Choose the most appropriate answer from the options given below:

(1) (b), (d) and (e) only

(2) (a) and (e) only

(3) (b) and (c) only

(4) (a) and (c) only

17. Given below are two statements : **[Re-NEET-2022]**

Assertion (A) : FSH which interacts with membrane bound receptors does not enter the target cell.

Reason (R) : Binding of FSH to its receptors generates second messenger (cyclic AMP) for its biochemical and physiological responses.

In the light of the above statements, choose the most appropriate answer from the options given below ;

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

18. Erythropoietin hormone which stimulates R.B.C. formation is produced by : **[NEET-2021]**

(1) Alpha cells of pancreas

(2) The cells of rostral adenohypophysis

(3) The cells of bone marrow

(4) Juxtaglomerular cells of the kidney

19. Which of the following acts as physiological barrier? **[NEET-2021]**

(1) Natural killer cells

(2) Interferons

(3) Tears from eyes

(4) Mucus coating of the epithelial lining of urogenital tracts

20. Select the correct statement. **[NEET-2020]**

(1) Insulin is associated with hyperglycemia

(2) Glucocorticoids stimulate gluconeogenesis

(3) Glucagon is associated with hypoglycemia.

(4) Insulin acts on pancreatic cells and adipocytes.

21. Match the following columns and select the correct option : **[NEET-2020]**

Column-I	Column-II
(a) Pituitary gland	(i) Grave's disease
(b) Thyroid gland	(ii) Diabetes mellitus
(c) Adrenal gland	(iii) Diabetes insipidus
(d) Pancreas	(iv) Addison's disease

	(a)	(b)	(c)	(d)
(1)	(ii)	(i)	(iv)	(iii)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(iii)	(i)	(iv)	(ii)

22. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus ? **[NEET-2020]**

(1) Renal calculi and Hyperglycaemia

(2) Uremia and Ketonuria

(3) Uremia and Renal Calculi

(4) Ketonuria and Glycosuria

23. Match the following columns and select the correct option :- **[NEET-2020 COVID-19]**

Column-I	Column-II
(a) Pituitary hormone	(i) Steroid
(b) Epinephrine	(ii) Neuropeptides
(c) Endorphins	(iii) Peptides, proteins
(d) Cortisol	(iv) Biogenic amines

(1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

24. Hormones stored and released from neurohypophysis are :- **[NEET-2020 COVID-19]**

(1) Thyroid stimulating hormone and Oxytocin

(2) Oxytocin and Vasopressin

(3) Follicle stimulating hormone and Leutinizing hormone

(4) Prolactin and Vasopressin

25. How does steroid hormone influence the cellular activities? **[NEET-2019]**

(1) Changing the permeability of the cell membrane.

(2) Binding to DNA and forming a gene hormone complex.

(3) Activating cyclic AMP located on the cell membrane.

(4) Using aquaporin channels as second messenger.

26. Match the following hormones with the respective disease : **[NEET-2019]**

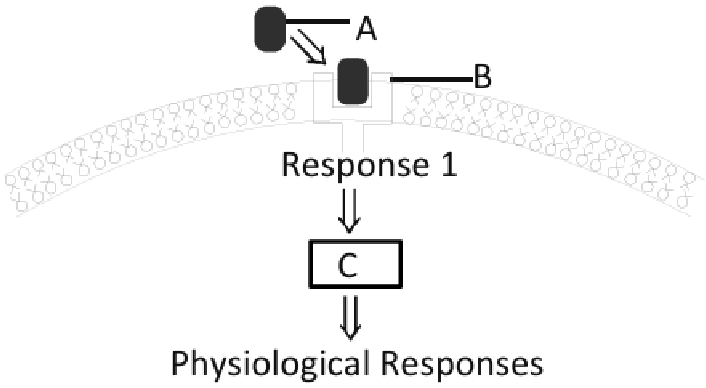
(a) Insulin	(i) Addison's disease
(b) Thyroxin	(ii) Diabetes insipidus
(c) Corticoids	(iii) Arcomegaly
(d) Growth Hormone	(iv) Goitre
	(v) Diabetes mellitus

Select the correct option.

	(a)	(b)	(c)	(d)
(1)	(v)	(i)	(ii)	(iii)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(v)	(iv)	(i)	(iii)
(4)	(ii)	(iv)	(i)	(iii)

27. Identify A, B and C in the diagramatic representation of the mechanism of hormone action.

[NEET-2019-ODISHA]



Select the correct option from the following :

(1) A-Steroid Hormone; B-Hormone receptor Complex, C-Protein

(2) A-Protein Hormone, B-Receptor; C-Cyclic AMP

(3) A-Steroid Hormone; B-Receptor, C - Second Messenger

(4) A-Protein Hormone; B-Cyclic AMP, C-Hormone-receptor Complex

28. Which of the following hormones is responsible for both the milk ejection reflex and the foetal ejection reflex ? **[NEET-2019-ODISHA]**

(1) Estrogen

(2) Prolactin

(3) Oxytocin

(4) Relaxin

29. Which of the following conditions will stimulate parathyroid gland to release parathyroid hormone? **[NEET-2019-ODISHA]**

(1) Fall in active Vitamin D levels

(2) Fall in blood Ca+2 levels

(3) Fall in bone Ca+2 levels

(4) Rise in blood Ca+2 levels

30. Artificial light, extended work-time and reduced sleep–time disrupt the activity of **[NEET-2019-ODISHA]**

(1) Thymus gland (2) Pineal gland

(3) Adrenal gland (4) Posterior pituitary gland

31. Which of the following is an amino acid derived hormone ? **[NEET-2018]**

(1) Epinephrine

(2) Ecdysone

(3) Estradiol

(4) Estriol

32. Which of the following hormones can play a significant role in osteoporosis ? **[NEET-2018]**

(1) Aldosterone and Prolactin

(2) Progesterone and Aldosterone

(3) Estrogen and Parathyroid hormone

(4) Parathyroid hormone and Prolactin

33. A temporary endocrine gland in the human body is: **[NEET-2017]**

(1) Corpus cardiacum

(2) corpus luteum

(3) Corpus allatum

(4) Pineal gland

34. GnRH, a hypothalamic hormone, needed in reproduction, acts on: **[NEET-2017]**

(1) anterior pituitary gland and stimulates secretion of LH and FSH.

(2) posterior pituitary gland and stimulates secretion of oxytocin and FSH.

(3) posterior pituitary gland and stimulates secretion of LH and relaxin.

(4) anterior pituitary gland and stimulates secretion of LH and oxytocin.

35. Hypersecretion of Growth Hormone in adults does not cause further increase in height, because:

(1) Epiphyseal plates close after adolescence. **[NEET-2017]**

(2) Bones loose their sensitivity to Growth Hormone in adults.

(3) Muscle fibres do not grow in size after birth.

(4) Growth Hormone becomes inactive in adults.

36. Which of the following pairs of hormones are not antagonistic (having opposite effects) to each other?

(1) Parathormone – Calcitonin (2) Insulin – Glucagon **[NEET- I 2016]**

(3) Aldosterone – Atrial Natriuretic Factor (4) Relaxin – Inhibin

37. Graves' disease is caused due to :- **[NEET- II 2016]**

(1) Hyposecretion of adrenal gland

(2) Hypersecretion of adrenal gland

(3) Hyposecretion of thyroid gland

(4) Hypersecretion of thyroid gland

38. Name a peptide hormone which acts mainly on hepatocytes, adipocytes and enhances cellular glucose uptake and utilization. **[NEET- II 2016]**

(1) Secretin

(2) Gastrin

(3) Insulin

(4) Glucagon

39. Osteoporosis, an age–related disease of skeletal system, may occur due to :- **[NEET- II 2016]**

(1) Decreased level of estrogen

(2) Accumulation of uric acid leading to inflammation of joints.

(3) Immune disorder affecting neuro muscular junction leading to fatigue.

(4) High concentration of Ca^{++} and Na^{+}.

40. The posterior pituitary gland is not a 'true' endocrine gland because :- **[NEET- II 2016]**

(1) It is under the regulation of hypothalamus

(2) It secretes enzymes

(3) It is provided with a duct

(4) It only stores and releases hormones

ANSWER KEY

1. (1)
2. (2)
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